Ta-Yang Wang

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Professional Summary

Software Engineer at Databricks and Ph.D. in Computer Science focused on building machine learning applications from research to production. Experienced in applying ML to production data applications at scale, with current work centered on forecasting-based anomaly detection for data quality. Research background in Graph Neural Networks (GNNs), large-scale model training acceleration, and distributed resource optimization.

Education

University of Southern California (USC) — Los Angeles, CA

Ph.D. in Computer Science — *Aug. 2019 – Oct. 2025*

- Dissertation — Heterogeneous Graph Representation Learning: Algorithms and Applications
- Reviewer: WWW 2025, NeurIPS 2024, FPGA 2022 & 2023, SC 2022, CLOUD 2022, HPEC 2021, CCGrid 2021

National Taiwan University (NTU) — Taipei, Taiwan

B.S. in Mathematics | Overall GPA: 3.96/4.3 | Rank: 2nd/55 — *Sep. 2014 – June 2018*

- Honorable Mention, 9th S.-T. Yau College Student Mathematics Contest
  - Named a finalist (top 15) in Applied and Computational Mathematics.
- National Taiwan University Presidential Award (Twice)
  - Awarded to students ranking in the top 5% of the department by semester GPA.

Experience

Databricks — Mountain View, CA

Software Engineer — *Oct. 2025 – Present*

- Build anomaly detection infrastructure for Unity Catalog Data Quality Monitoring, helping teams trust business-critical analytics, dashboards, and ML pipelines by continuously evaluating table freshness and completeness.
- Develop per-table forecasting models that learn historical commit and row-volume patterns to predict next commit time and expected 24-hour row-count ranges.
- Surface predicted-versus-observed quality signals and health indicators, enabling users to detect stale or incomplete tables, prioritize incidents, and investigate data quality issues at scale.

Data Science Lab, University of Southern California — Los Angeles, CA

Graduate Research Assistant — *Aug. 2019 – Oct. 2025*

- Developed bandit-sampling algorithms for real-world graph representation learning, addressing missing node attributes in Heterogeneous Graph Neural Networks (HGNNs).
- Accelerated large-scale HGNN training by 50% while maintaining state-of-the-art (SOTA) accuracy, with a theoretical convergence guarantee for adaptive sampling.

Medical Data Analytics Lab, National Center for Theoretical Sciences — Taipei, Taiwan

Research Assistant — *Sep. 2018 – June 2019*

- Collaborated with National Taiwan University Hospital on medical imaging models for clinical applications.
- Designed a persistent-homology-based model for prostate cancer histopathology image classification.

Taiwan Chapter of the Society of American Baseball Research (SABR) — Taipei, Taiwan

Co-founder — *Sep. 2017 – June 2018*

- Architected an end-to-end pipeline to ingest and analyze large-scale TrackMan datasets, delivering a quantitative player evaluation system for Taiwan's Professional Baseball League (CPBL).
- Developed predictive models from StatCast and PITCHf/x data to create objective, data-driven pitch-quality metrics for Major League Baseball (MLB) player assessment.

Skills

Programming Languages: Python, C/C++, R, Matlab, SQL
Machine Learning & Data: Time Series Forecasting, Anomaly Detection, Graph Neural Networks, Bandit Algorithms, Data Quality Monitoring
Frameworks & Libraries: PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, SciPy
Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm

Publications

Graph Neural Networks

- **Ta-Yang Wang**, Rajgopal Kannan, and Viktor Prasanna. "Effective and Generalizable Pre-Trained Heterogeneous Graph Neural Networks with Bandit Samplers." [Under Review]
- **Ta-Yang Wang**, Rajgopal Kannan, and Viktor Prasanna. "TypeBandit: Type-Level Context Allocation and Reweighting for Effective Attribute Completion in Heterogeneous Graph Neural Networks." *IEEE Transactions on Big Data.* [Under Review]
- **Ta-Yang Wang**, Rajgopal Kannan, and Viktor Prasanna. "Heterogeneous Graph Neural Network based on Bandit Sampling for Attribute Completion." *The 15th IEEE International Conference on Knowledge Graphs (ICKG)*, 2024.
- **Ta-Yang Wang**, Rajgopal Kannan, and Viktor Prasanna. "Training Heterogeneous Graph Neural Networks using Bandit Sampling." *The 32nd ACM International Conference on Information and Knowledge Management (CIKM)*, 2023.
- **Ta-Yang Wang**, Hongkuan Zhou, Rajgopal Kannan, Ananthram Swami, and Viktor Prasanna. "Throughput Optimization in Heterogeneous MIMO Networks: A GNN-based Approach." *The 1st Graph Neural Networking Workshop (GNNet)*, 2022.

Task Mapping

- **Ta-Yang Wang**, William Chang, Ajitesh Srivastava, Rajgopal Kannan, and Viktor Prasanna. "Monte Carlo Tree Search for Task Mapping onto Heterogeneous Platforms." *The 28th IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC)*, 2021.
- **Ta-Yang Wang**, Ajitesh Srivastava, and Viktor Prasanna. "A Framework for Task Mapping onto Heterogeneous Platforms." *IEEE High Performance Extreme Computing Conference (HPEC)*, 2020.

Tensor Decomposition

- Sasindu Wijeratne, **Ta-Yang Wang**, Rajgopal Kannan, and Viktor Prasanna. "Accelerating sparse MTTKRP for tensor decomposition on FPGA." *The 32nd ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA)*, 2023.
- Sasindu Wijeratne, **Ta-Yang Wang**, Rajgopal Kannan, and Viktor Prasanna. "Towards Programmable Memory Controller for Tensor Decomposition." *The 11th International Conference on Data Science, Technology and Applications (DATA)*, 2022.

Memory Access Prediction

- Pengmiao Zhang, Ajitesh Srivastava, **Ta-Yang Wang**, Cesar AF De Rose, Rajgopal Kannan, and Viktor Prasanna. "C-MemMAP: Clustering-driven Compact, Adaptable, and Generalizable Meta-LSTM Models for Memory Access Prediction." *International Journal of Data Science and Analytics (JDSA)*, 2021.
- Ajitesh Srivastava, **Ta-Yang Wang**, Pengmiao Zhang, Cesar Augusto F De Rose, Rajgopal Kannan, and Viktor Prasanna. "MemMAP: Compact and Generalizable Meta-LSTM Models for Memory Access Prediction." *The 24th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 2020.